

Mobile Controlled Light

ESP32 + Wokwi + Blynk

Follow the steps in order. You need no hardware, only a browser and the free Blynk app.

Part 1: Set up Blynk first

1. Go to **blynk.cloud** and create a free account.

2. Open **Developer Zone** → **My Templates** → **New Template**.

Name: `Mobile Light` | Hardware: `ESP32` | Connection type: `WiFi`
Click **Done**.

3. Open the **Datastreams** tab → **New Datastream** → **Virtual Pin**.

Name: `Light` | Pin: `V0` | Data type: `Integer` | Min: `0` | Max: `1`
Click **Create**.

4. Open the **Web Dashboard** tab. Drag a **Switch** widget onto the canvas, open its settings, set the datastream to **Light (V0)**, and click **Save**.

5. Click **Save** on the template (top right).

6. Go to **Devices** → **New Device** → **From template**. Choose `Mobile Light`, name it `ESP32-Light`, and click **Create**.

7. On the device page, open the **Device Info** tab and copy these three values:

`BLYNK_TEMPLATE_ID`

`BLYNK_TEMPLATE_NAME`

`BLYNK_AUTH_TOKEN`

Keep them somewhere safe. You paste them into the code in Part 3.

Part 2: Build the circuit in Wokwi

1. Go to **wokwi.com** and sign in.

2. Click **New Project** → **ESP32**.

3. Click the blue **+** button (Add part) and add: **LED** (any colour) and **Resistor** (set its value to **220 Ω**).

4. Wire it exactly like this:

From	To
ESP32 GPIO 26	Resistor, pin 1
Resistor, pin 2	LED anode (the longer, bent leg)
LED cathode (the shorter leg)	ESP32 GND

The resistor protects the LED from too much current.

Part 3: Add the library and code

1. In Wokwi, click the **Library Manager** tab, search for **Blynk**, and add it.

2. Open `sketch.ino`, delete everything in it, and paste this code:

```
#define BLYNK_TEMPLATE_ID "PASTE_YOUR_TEMPLATE_ID"
#define BLYNK_TEMPLATE_NAME "PASTE_YOUR_TEMPLATE_NAME"
#define BLYNK_AUTH_TOKEN "PASTE_YOUR_AUTH_TOKEN"
```

```
#include <WiFi.h> #include
<BlynkSimpleEsp32.h>

char ssid[] = "Wokwi-GUEST"; // Wokwi's built-in virtual
WiFi char pass[] = ""; // no password #define
LED_PIN 26

// Runs automatically whenever the Blynk switch (V0)
changes BLYNK_WRITE(V0) { int value = param.asInt();
// 1 = ON, 0 = OFF digitalWrite(LED_PIN, value); }

void setup() { Serial.begin(115200);
pinMode(LED_PIN, OUTPUT);
digitalWrite(LED_PIN, LOW);
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
}

void loop() {
Blynk.run();
}
```

3. Replace the three `PASTE_YOUR_...` values with the ones you copied in Part 1.

Part 4: Run and test

1. Click the green **Start Simulation** button in Wokwi.
2. Watch the **Serial Monitor**. It should connect to Blynk within a few seconds.
3. In the Blynk web dashboard, flip the **Switch**. The LED in Wokwi should turn **ON**, and flipping it back turns it **OFF**.

Part 5: Control it from your phone

1. Install the **Blynk IoT** app (Android or iOS) and log in with the same account.
2. Open your device `ESP32-Light`.
3. Add a **Switch** widget to its dashboard and set it to datastream **Light (V0)**.
4. Toggle it, and the LED in Wokwi follows.

Keep the Wokwi simulation running while you test, because the “device” is only online while it runs.

If something goes wrong

Problem	Fix
Stuck on “Connecting to WiFi”	Make sure the SSID is exactly Wokwi-GUEST and the password is empty.
Blynk says device offline	Check that the Template ID, Name, and Auth Token in the code match Device Info exactly.
LED never lights	Check the LED direction (the longer leg goes toward the resistor) and that the wire goes to 26.
Switch does nothing	Confirm the datastream pin is V0 and the widget is bound to it.

Note: The Blynk website and app menus change from time to time, so button names may look slightly different. The order of steps stays the same.

